

DISCRETE FOURIER TRANSFORM (DFT)

DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- The discrete-time Fourier transform of the sequence as

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n}$$

- However, ω is a continuous variable taking value in $[0, 2\pi]$. The final step is to evaluate $X(e^{j\omega})$ at only a finite number of ω by sampling uniformly in the range $[0, 2\pi]$. We obtain

$$X(e^{j\omega_k}) = \sum_{n=0}^{N-1} x[n] e^{-j\omega_k n}, \text{ where } k = 0, 1, \dots, M-1 \quad (4.18)$$

- Where $\omega_k = \frac{2\pi}{M}k$. The number of frequency samples, M , can be any value. However, we choose it to be the same as the number of time samples N . With this modification we wrote $X[e^{j\omega_k}] = X[k]$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)nk} \quad (4.19)$$

- $X[k]$ is discrete Fourier transform of $x[n]$
- The inverse discrete Fourier-transform (IDFT) relation is given by

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)nk} \quad (4.20)$$

DISCRETE FOURIER TRANSFORM (DFT)

- To derive this relation, replace n with p in the right side of (4.19) and multiply by $e^{j(2\pi/N)nk}$ to get

$$X[k]e^{j(2\pi/N)nk} = \sum_{p=0}^{N-1} x[p]e^{-j(2\pi/N)(p-n)k}$$

- Now sum over k in the range $[0, N-1]$ to get

$$\sum_{k=0}^{N-1} X[k]e^{j(2\pi/N)nk} = \sum_{k=0}^{N-1} \sum_{p=0}^{N-1} x[p]e^{-j(2\pi/N)(p-n)k}$$

- From (4.7)

$$\sum_{k=0}^{N-1} e^{-j(2\pi/N)(n-p)k} = \begin{cases} N, & n = p \\ 0, & n \neq p \end{cases} \quad (4.21)$$

- From $X(e^{j\omega})$ is periodic in ω with period 2π , thus $X(e^{j\omega_k}) = X(e^{j\omega_k+2\pi})$. This can be written as

$$X[k] = X(e^{j\omega}) = X(e^{j\omega+2\pi}) = X\left(\frac{2\pi}{N}(k+N)\right) = X[k+N] \quad (4.22)$$

- That is, $X[k]$ is periodic with period N . $x[n]$ is also periodic with period N .

$$\begin{aligned} x[n+N] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{j(2\pi/N)(n+N)k} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{j(2\pi/N)(n)k} \\ &= x[n] \end{aligned} \quad (4.23)$$

DISCRETE FOURIER TRANSFORM (DFT)

PROPERTIES

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)nk}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)nk}$$

(1) Linearity

Let $X_1[k]$ and $X_2[k]$ be the DFTs of the two sequences $x_1[n]$ and $x_2[n]$, then

$$\text{DFT}[a_1 x_1[n] + a_2 x_2[n]] = a_1 X_1[k] + a_2 X_2[k] \text{ for any constants } a_1 \text{ and } a_2 \quad (4.24)$$

(2) Time shifting / Frequency shifting

For any integer n_0 ,

$$\begin{aligned} \text{DFT}[x[n + n_0]] &= \sum_{n=0}^{N-1} x[n + n_0] e^{-j(2\pi/N)(n)k} \\ &= \sum_{\langle N \rangle} x[m] e^{-j(2\pi/N)(m-n_0)k} = e^{j(2\pi/N)n_0 k} X[k] \end{aligned} \quad (4.25)$$

The shift is a circular shift.

For any integer k_0 ,

$$\begin{aligned} \text{IDFT}[X[k + k_0]] &= \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)n(k+k_0)} \\ &= \sum_{\langle N \rangle} X[m] e^{j(2\pi/N)n(m-k_0)} = e^{-j(2\pi/N)nk_0} x[n] \end{aligned} \quad (4.26)$$

DISCRETE FOURIER TRANSFORM (DFT)

PROPERTIES

(3) Time Convolution

The inverse transform of the product of two transforms corresponds to a convolution of the corresponding time functions. With this in view, let us determine IDFT of the function

$$Y[k] = H[k]X[k]$$

$$\begin{aligned} y[n] = IDFT[Y[k]] &= \frac{1}{N} \sum_{k=0}^{N-1} Y[k] e^{j(2\pi/N)(n)k} = \frac{1}{N} \sum_{k=0}^{N-1} H[k] X[k] e^{j(2\pi/N)(n)k} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} \sum_{m=0}^{N-1} \underbrace{h[m] e^{-j(2\pi/N)mk} X[k] e^{j(2\pi/N)nk}}_{x[n-m]} = \sum_{m=0}^{N-1} h[m] x[n-m] = h[n] * x[n] \end{aligned} \quad (4.27)$$

A comparison with Lecture 02 Convolution shows that the right hand side of (4.27) corresponds to the periodic convolution of the two sequences $x[n]$ and $h[n]$

DISCRETE FOURIER TRANSFORM

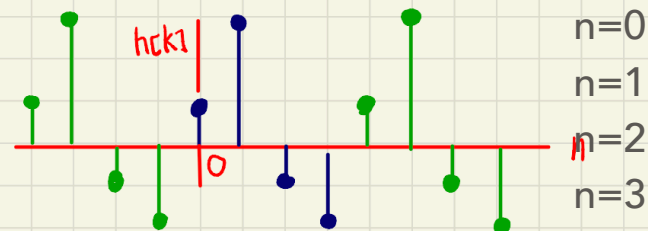
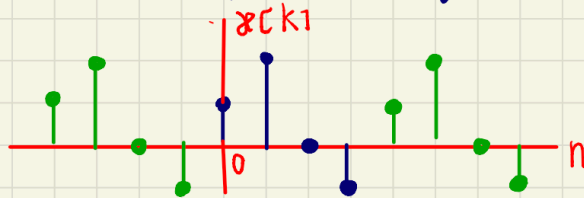
EXERCISE #3.3.1 CONVOLUTION



Find the convolution of the periodic extensions of the two sequences.

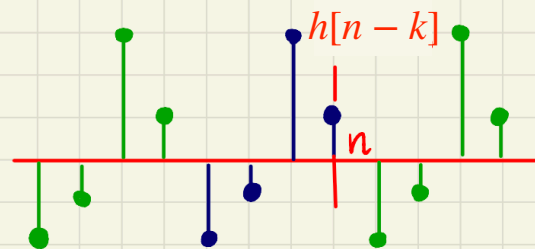
Time-Domain

$$x[n] = \{1, 2, 0, -1\} \quad \text{and} \quad h[n] = \{1, 3, -1, -2\}$$



Periodic Convolution

$x[k]$	1	2	0	-1	
$n \ k$	0	1	2	3	
0 $h[-k]$	1	-2	-1	3	
1 $h[1-k]$	3	1	-2	-1	
2 $h[2-k]$	-1	3	1	-2	
3 $h[3-k]$	-2	-1	3	1	
0 $h[-k]x[k]$	1	-4	0	-3	$y[0] = -6$
1 $h[1-k]x[k]$	3	2	0	1	$y[1] = 6$
2 $h[2-k]x[k]$	-1	6	0	2	$y[2] = 7$
3 $h[3-k]x[k]$	-2	-2	0	-1	$y[3] = -5$



DISCRETE FOURIER TRANSFORM

Find the convolution of the periodic extensions of the two sequences.

$$x[n] = \{1, 2, 0, -1\} \quad \text{and} \quad h[n] = \{1, 3, -1, -2\}$$

using DFT, $N=4$ so that $e^{j\frac{2\pi}{N}n} = e^{j\frac{2\pi}{4}n} = e^{j\frac{\pi}{2}n} = j$ $\therefore e^{-j\frac{2\pi}{N}nk} = (j)^{-nk}$

Calculate the DFTs of $x[n]$ and $h[n]$

$$\text{From } X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk},$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}nk}$$

$$H[0] = h[0] + h[1] + h[2] + h[3] = 1$$

$$H[1] = h[0] + h[1](j)^{-1} + h[2](j)^{-2} + h[3](j)^{-3} = 1 - 3j + 1 - 2j = 2 - 5j$$

$$H[2] = h[0] + h[1](j)^{-2} + h[2](j)^{-4} + h[3](j)^{-6} = 1 - 3 - 1 + 2 = -1$$

$$H[3] = h[0] + h[1](j)^{-3} + h[2](j)^{-6} + h[3](j)^{-9} = 1 + 3j + 1 + 2j = 2 + 5j$$

Freq-
Domain

$$x[n] = \{1, 2, 0, -1\} \quad e^{j\frac{2\pi}{N}nk} = (j)^{nk}$$

Freq-
Domain

$$X[0] = x[0] + x[1] + x[2] + x[3] = 1 + 2 + 0 - 1 = 2$$

$$X[1] = x[0] + x[1](j)^{-1} + x[2](j)^{-2} + x[3](j)^{-3} = 1 - 2j + 0 - 1j = 1 - 3j$$

$$X[2] = x[0] + x[1](j)^{-2} + x[2](j)^{-4} + x[3](j)^{-6} = 1 - 2 + 0 + 1 = 0$$

$$X[3] = x[0] + x[1](j)^{-3} + x[2](j)^{-6} + x[3](j)^{-9} = 1 + 2j + 0 + 1j = 1 + 3j$$

$$Y[0] = X[0] H[0] = (1)(2) = 2$$

$$Y[1] = X[1] H[1] = (2 - 5j)(1 - 3j) = -13 - 11j$$

$$Y[2] = X[2] H[2] = (-1)(0) = 0$$

$$Y[3] = X[3] H[3] = (2 + 5j)(1 + 3j) = -13 + 11j$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}nk}$$

$$Y[k] = \{2, -13 - 11j, 0, -13 + 11j\}$$

$$N=4, \quad e^{j\frac{2\pi}{N}nk} = (j)^{nk}$$

Freq-
Domain

$$y[0] = \frac{1}{4} [Y[0] + Y[1] + Y[2] + Y[3]] = \frac{1}{4} [2 - 13 - 11j + 0 - 13 + 11j] = -6$$

$$y[1] = \frac{1}{4} [Y[0] + Y[1](j)^{-1} + Y[2](j)^{-2} + Y[3](j)^{-3}] = \frac{1}{4} [2 - 13j + 11 + 0 + 13j + 11] = 6$$

$$y[2] = \frac{1}{4} [Y[0] + Y[1](j)^{-2} + Y[2](j)^{-4} + Y[3](j)^{-6}] = \frac{1}{4} [2 + 13 + 11j + 0 + 13 - 11j] = 7$$

$$y[3] = \frac{1}{4} [Y[0] + Y[1](j)^{-3} + Y[2](j)^{-6} + Y[3](j)^{-9}] = \frac{1}{4} [2 + 13j - 11 + 0 - 13j - 11] = -5$$

$$y[n] = \{-6, 6, 7, -5\}$$

DISCRETE FOURIER TRANSFORM (DFT) RELATIONSHIP

(1) Relationship between the DFT and Fourier series for Discrete-time Signals

$x[n]$ can be interpreted as the coefficients a_k in Fourier series for Discrete-time Signals with periodic extension multiplied by period N , that is

$$X[k] = Na_k \quad (4.28)$$

(2) Relationship between DFT and Fourier transform for Discrete-time Signals

From (4.15), Fourier transform of $x[n]$ can be expressed as

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (4.29)$$

$$\text{We saw that } X[k] = X(e^{j\omega})|_{\omega=(2\pi/N)k} = X\left(\frac{2\pi}{N}k\right)$$

$X[k]$ corresponds to the sampled $X(e^{j\omega})$ at the uniformly spaced frequencies $\omega = \frac{2\pi}{N}k$ for integer k .

Fourier series

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N} \sum_{k=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$$

Fourier transform

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

DFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)nk}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)nk}$$

DISCRETE FOURIER TRANSFORM (DFT)

MATRIX INTERPRETATION OF DFT

DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)nk}$$

- We can express the DFT relation of (4.19) compactly as a matrix operation on the data vector $x = [x(0) \ x(1) \ \dots \ x(N-1)]^T$. Let us denote $e^{-j2\pi/N}$ by W_N . We can then write

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}, \quad k = 0, 1, \dots, N-1 \quad (4.30)$$

- Let W be the matrix whose (k,n) th element $[W]_{kn}$ is equal to W_N^{nk} . That is,

$$W = \begin{bmatrix} W_N^0 & W_N^0 & W_N^0 & \cdot & W_N^0 \\ W_N^0 & W_N^1 & W_N^2 & \cdot & W_N^{N-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ W_N^0 & W_N^{N-1} & W_N^{2(N-1)} & \cdot & W_N^{(N-1)(N-1)} \end{bmatrix} \quad (4.31)$$

DISCRETE FOURIER TRANSFORM (DFT)

MATRIX INTERPRETATION OF DFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)nk}$$

The transform vector $X = [X(0) X(1) \dots X(N-1)]^T$ can be obtained as

$$X = Wx \quad (4.32)$$

W referred as DFT matrix. Clearly, $[W]_{kn} = [W]_{nk}$, so that W is symmetric ($W = W^T$). We can also write

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-nk} \quad (4.33)$$

Since $W_N^{-1} = W_N^*$, IDFT can be written in matrix form as

$$x = \frac{1}{N} W^* X, \text{ then from (4.32) we will get, } x = W^{-1} X \quad (4.34)$$

$$\text{So we have } W^{-1} = \frac{1}{N} W^* \text{ or } W^* W = N I_N \quad (4.35)$$

DISCRETE FOURIER TRANSFORM (DFT)

N-point sequence $x[n] \leftrightarrow X[k]$

- Rectangular pulse with $N = 11$ points, $N_1 = 5$
- Rectangular pulse with $N = 21$ points, $N_1 = 11$
- $N = 21$ points



FAST FOURIER TRANSFORM (FFT)

FAST ALGORITHM FOR DFT

Recall the DFT of the sequence $x[n]$ is given by

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}, \quad k = 0, 1, \dots, N-1 \quad (4.36)$$

Which can be explicitly written as

$$X[k] = x[0]W_N^0 + x[1]W_N^k + x[2]W_N^{2k} + \dots + x[N-1]W_N^{(N-1)k} \quad (4.37)$$

The direct determination of the DFT requires N^2 multiplications and N^2 additions ($O(N^2)$ Operations). Fast Fourier-transform (FFT) algorithm is a procedure to reduce the computational burden. The basic ideal is to divide the given sequence into subsequences of smaller length. Then combine these smaller DFTs suitably to obtain the DFT of the original sequence

The **Decimation-in-Time algorithm** divide $x[n]$ into two subsequences, each of length $N/2$ (grouping even-indexed samples and the odd-indexed samples together. We can write

$$X[k] = \sum_{n \text{ even}} x[n] W_N^{nk} + \sum_{n \text{ odd}} x[n] W_N^{nk} \quad (4.38)$$

FAST FOURIER TRANSFORM (FFT)

FAST ALGORITHM FOR DFT

Letting $n = 2r$ in the first sum and $n = 2r + 1$ in the second sum, we can write

$$X[k] = \sum_{r=0}^{N/2-1} x[2r]W_N^{2rk} + \sum_{r=0}^{N/2-1} x[2r+1]W_N^{(2r+1)k} \quad (4.39)$$

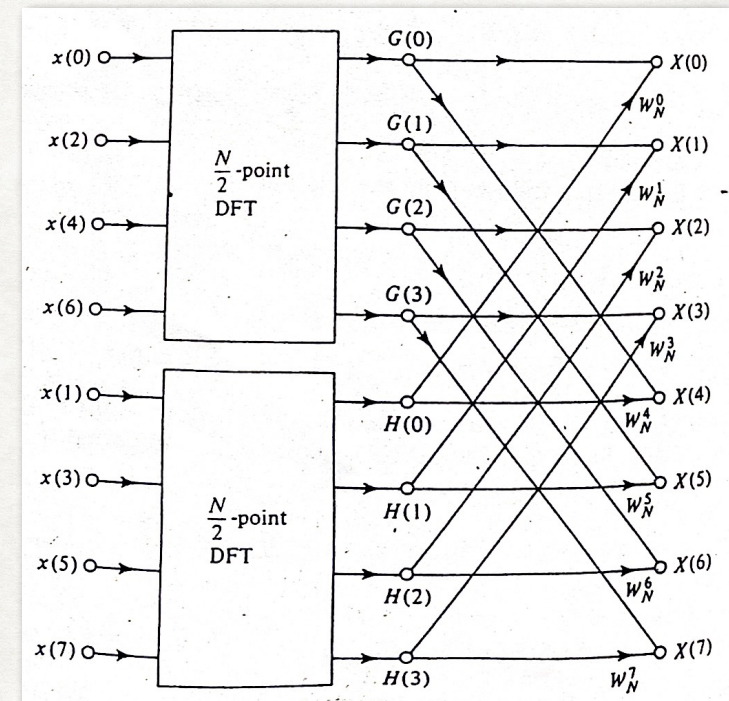
$$= \sum_{r=0}^{N/2-1} g[r]W_N^{2rk} + W_N^k \sum_{r=0}^{N/2-1} h[r]W_N^{2rk} \quad \text{where } g[r] = x[2r] \text{ and } h[r] = x[2r+1]$$

$$= G[k] + W_N^k H[k] \quad \text{for } k = 0, 1, \dots, N-1 \quad (4.40)$$

$G[k]$ and $H[k]$ denote the $N/2$ -point DFTs of the sequences $g[r]$ and $h[r]$, respectively.

Since $G[k]$ and $H[k]$ is periodic with period $N/2$: $X[k + \frac{N}{2}] = G[k] + W_N^{k+\frac{N}{2}} H[k]$

Each of the two DFTs requires two $(N/2)^2$ and N complex multiplications and two $(N/2)^2$ and N complex additions; total computation requires $N + N^2/2$ complex additions and multiplications.



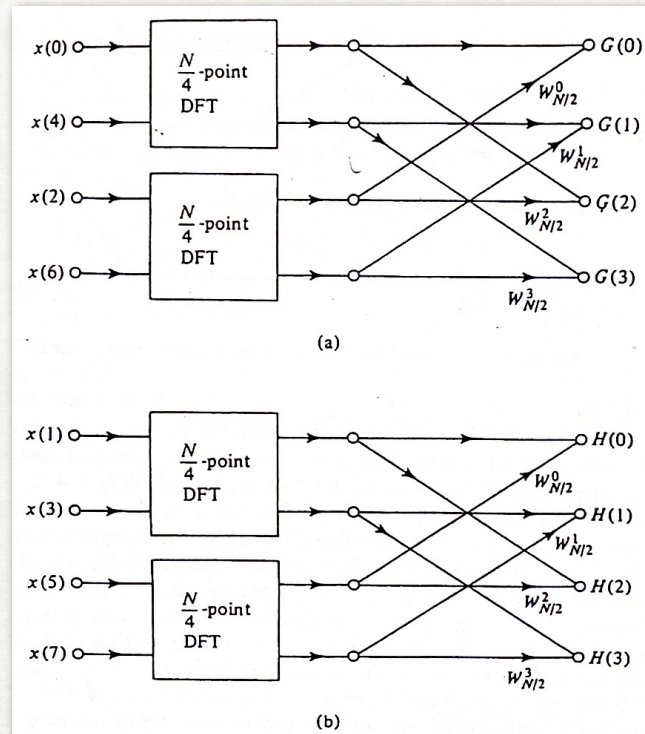
SIGNAL FLOW GRAPH FOR 2-POINT DFT FOR 8-POINT SEQUENCE

FAST FOURIER TRANSFORM (FFT)

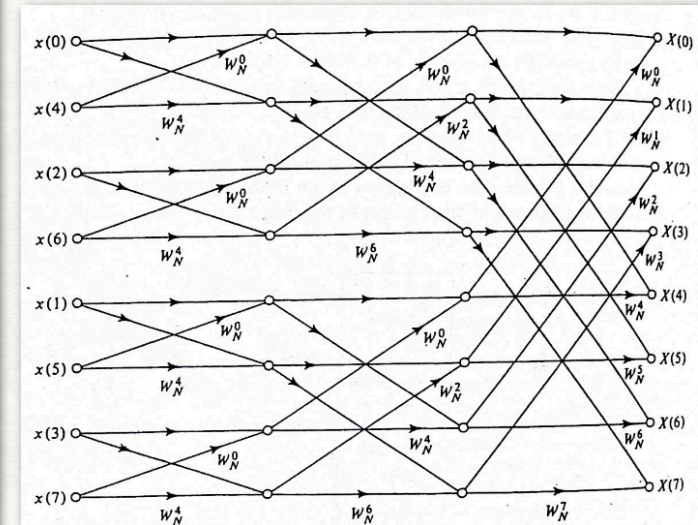
FAST ALGORITHM FOR DFT

- Determining $N/4$ -point DFTs, computation complexity for 4-point DFTs is

$$2\left(2\left(\frac{N}{4}\right)^2 + \frac{N}{2}\right) + N = \frac{N^2}{4} + 2N$$
- The procedure can be continued by further subdividing the subsequences until we get a set of two-point sequences.
- The complete graph flow for the computation of an eight-point DFT is shown in figure on the right-hand side.
- For 2^P -point DFTs, computation complexity is $N + N(\log_2 N) \rightarrow O(N(\log_2 N))$



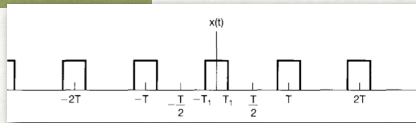
SIGNAL FLOW GRAPH FOR 4-POINT DFT



SIGNAL FLOW GRAPH FOR 8-POINT DFT

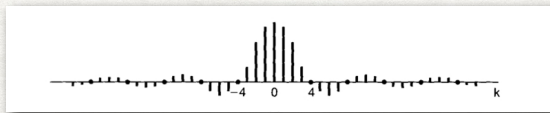
COMPARISON

FOURIER SERIES

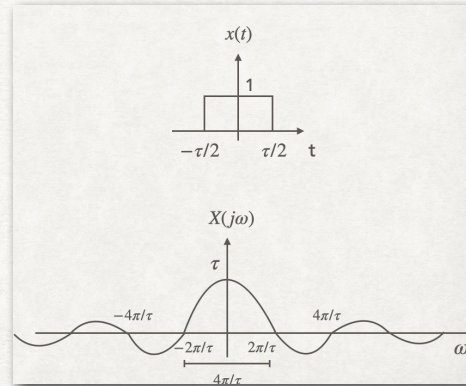


$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_0^T x(t) e^{-jk(2\pi/T)t} dt$$



FOURIER TRANSFORM



$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

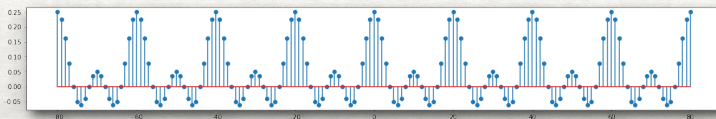
CONTINUOUS-TIME

FOURIER SERIES



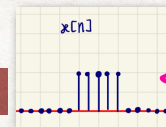
$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk(\omega_0)n} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n}$$



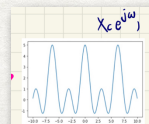
DISCRETE-TIME

FOURIER TRANSFORM



$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

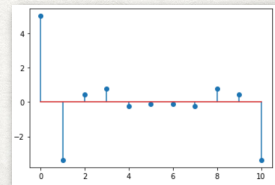
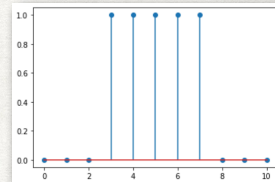
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$



DFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)nk}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)nk}$$



REFERENCES

- Samir S. Soliman and Mandyam D. Srinath., **Continuous and Discrete Signals and Systems**, 2nd Edition, Prentice Hall (1998)
- Chapter 9